Labeling

Amazon ⇒ 1
Google ⇒ 2
TCS ⇒ 3
Micr ⇒ 4
⋮ ⇒ 5
⋮
⇒ 10

[Multi-class classification]

10 : comp

Predic

DSA	Proje	IQ	Atte	CGPA	Goes ⇒ Output
200	4	110	80	7.8	1
					2
					4
					3

$w_1 x + w_2 x y + w_3 z + w_4 * a + w_5 x c + b \Rightarrow$ Output

10

Amazon
Google
TCS
Micr

① Sabse jaad chance kuse company select

TCS Mic Am . . . Goo

10 ⇒

[0.2 0.3 0.1 0.15 0.16 0.20]

Sum

30% chance Microsoft

Multilabel ⇒ Classification

1. 90% : Microsoft
2. 70% Google
3. 74% TCS
4. 85% Uber

⋮

10. 85%

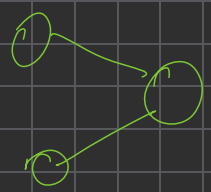
DSA	IO	Project	tra data
—	—	—	1
—	—	—	0

Train Kiyehai

1. ⇒ Model ⇒ Train ! ✓ Microsoft
2. ⇒ Model ⇒
3.

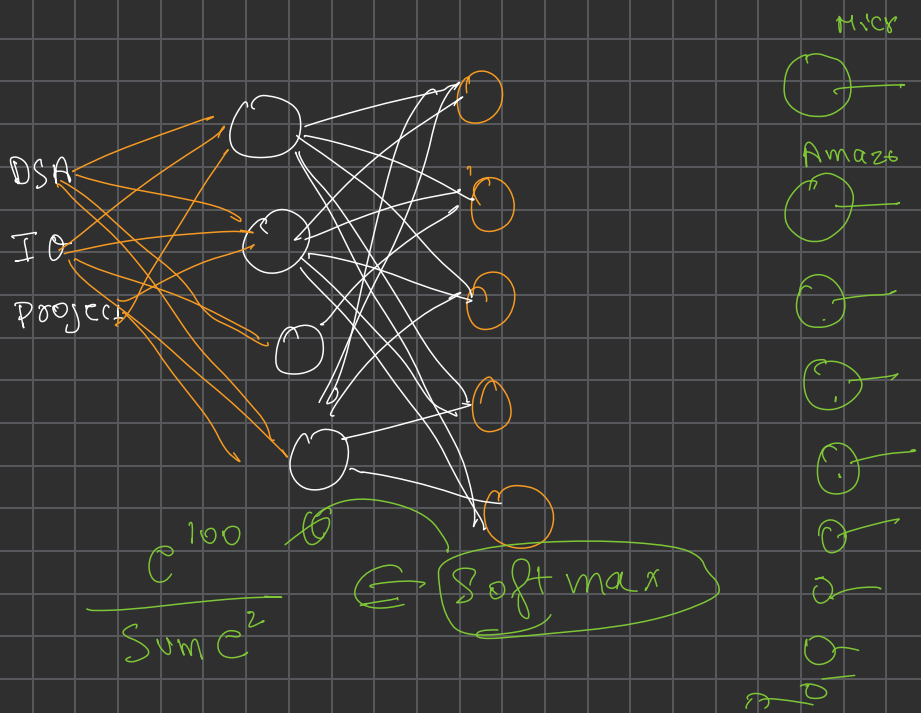
⋮

10



Data

sle.	padai	Moru
2	3	17
3	5	28
7	11	80
⋮		



$$\frac{e^{100}}{\text{Sum } e^{z}} \Leftarrow \text{Softmax}$$

e^{100} e^{50} e^{80} e^{-30} Codin = 0.4

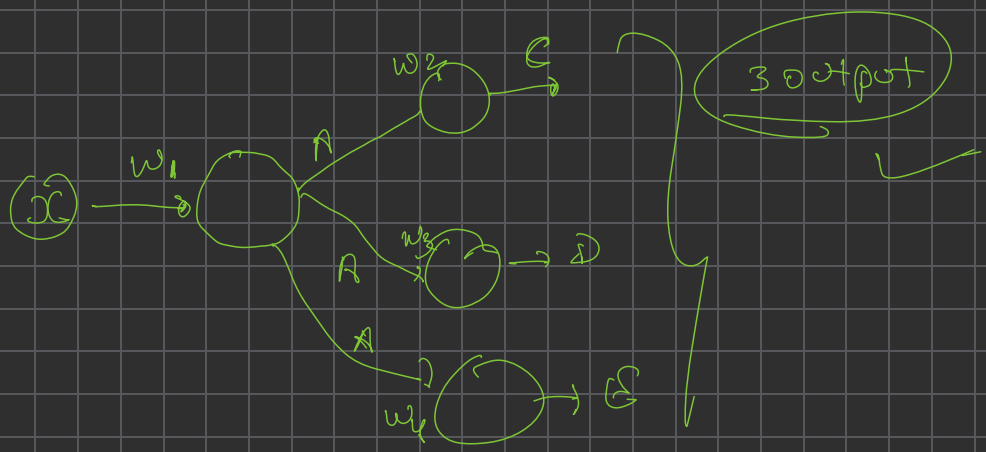
I love — Te = 0.3

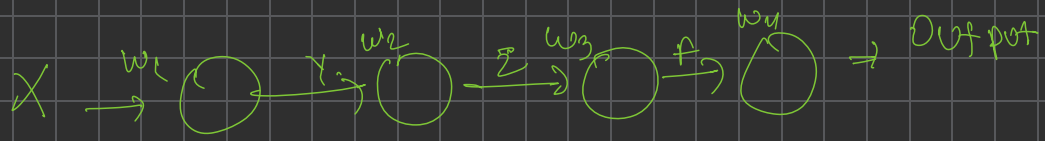
[100, 50, 8.0, -30, 20, 9.0, 7.5, -100, 20, 5]

[0.1, 0.2, 0.01, 0.5 ...]

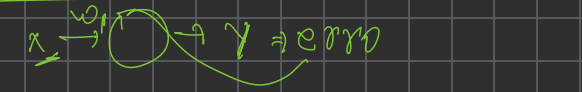
+ve

sum





$$\boxed{w_{1new} = w_{1old} + \eta * error * x}$$



Micro Amazon Google TCS . . .

[0.1, 0.4, 0.2, 0.1, 0.1, 0.01, 0.01, 0.03, 0.02, 0.02]

[2, 3, 4, 1]

⇓

[0.2 0.3 0.4 0.1]

11

1

$\frac{2}{10}$

[-1, 4, 2, 3]

be kar method

[0, 11, 9, 10]

[0, 0.33, 0.28, 0.3]

$\frac{11}{30}$

$$[-100000,\ 10,\ 1000]$$

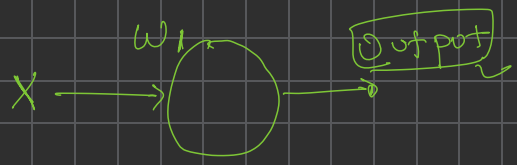
$$[0,\ 100010,\ 101000]$$

$$\Rightarrow [0,\ 0.49,\ 0.51]$$

Study	Marks
3	30
4	40
5	50

Actual oput
- predic

$\boxed{\text{error}} \Rightarrow$

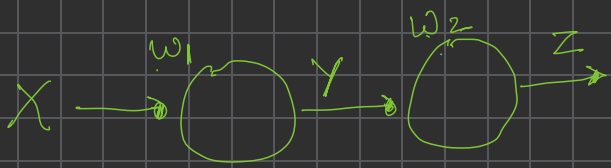


Output = $w_1 * x$

$w_{1n} = w_{1old} + \eta * \underline{error * x}$

error = Act - Z

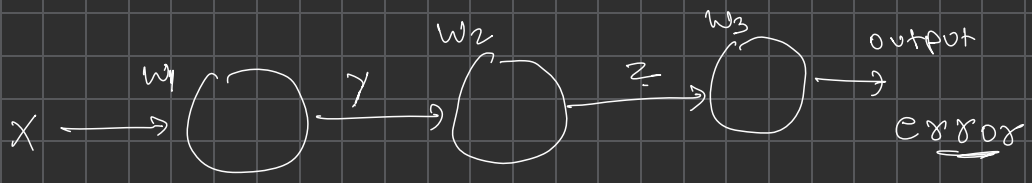
error



$error_2 = \boxed{w_1 * x}$

$$\left[w_{2new} = w_{2old} + \eta * error * Y \right]$$

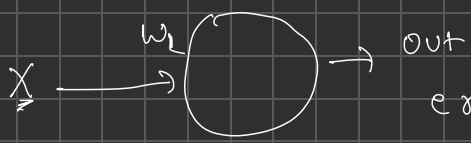
$$\left[w_{1new} = w_{1old} + \eta * \underbrace{\{error * w_2\}}_{\boxed{error}} * X \right]$$



$$w_{3new} = w_{3old} + \eta * error * z$$

$$w_{2new} = w_{2old} + \eta * (error * w_3) * y$$

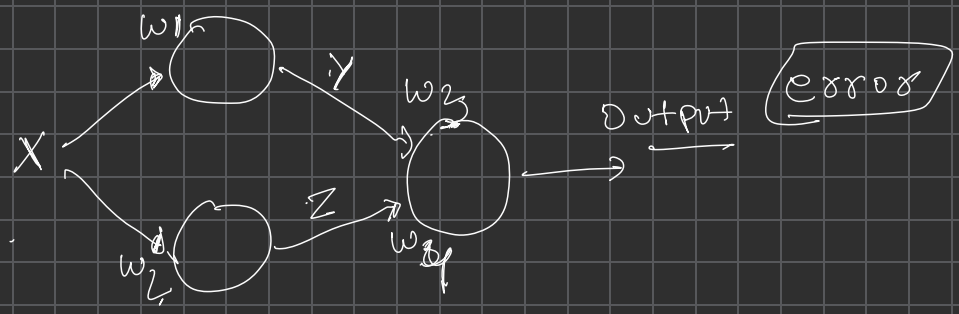
$$w_{1new} = w_{1old} + \eta * (error * w_3 * w_2) * x$$



$$w_{1new} = [w_{1old} + \eta * error * x]$$

$$w_{1new} = w_{1old} + \eta(error \times w_3) * x$$

$$w_{2new} = w_{2old} + \eta(error * w_y) * x$$



$$w_{3new} = w_{3old} + \eta * error \times y$$

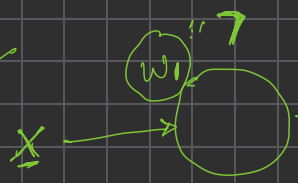
$$w_{ynew} = w_{yold} + \eta * error \times z$$

$y = \boxed{10x}$

$y \Rightarrow \boxed{7x}$

Stud	Mar
2	20
3	30
4	40

$w_{Avg} \Rightarrow 10$



Output

$\boxed{w_1 * x}$

$$\left[w_{1new} \Rightarrow w_{1old} + \eta * \underline{error} \times x \right]$$

error

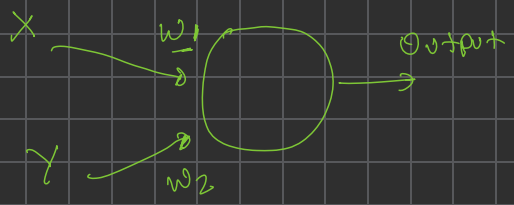
$$w_1 \Rightarrow \left(w_{Actual} + w_{error} \right) * x$$

$$7 \rightarrow \underbrace{10 - 3}\} + w_{Actual} x + \boxed{w_{cor} * x}$$

Correction

$$error \Rightarrow \boxed{w_{error} * x}$$

$$Output = w_1 * x$$



$$\text{Output} \Rightarrow w_1 * x + w_2 * y$$

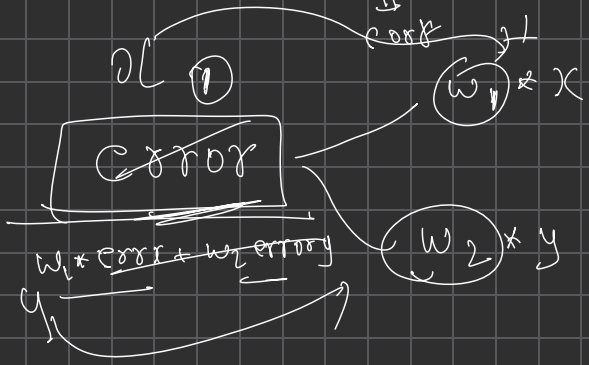
$$\Rightarrow (w_{1Actual} + w_{1error}) * x + (w_{2Actual} + w_{2error}) * y$$

$$\Rightarrow \left[w_{1Actual} * x + w_{2Actual} * y\right] + \left[w_{1error} * x + w_{2error} * y\right]$$

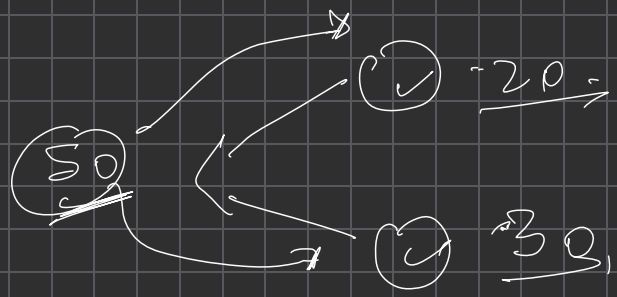
$$\Downarrow$$

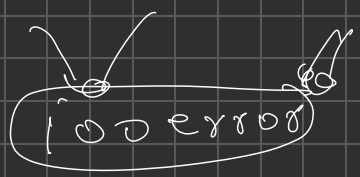
$$\text{corr}$$

$$\text{Error}$$



$$w_{1\,old} = w_{1\,new} + \eta * \text{error} * x$$



20 30 50

2 : 3 : 5